Federated Learning for Network Intrusion Classification: FedAvg Performance and Resilience to Label Poisoning on CIC–UNSW-NB15

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Independent Research / Academic Project

***Abstract*—** Federated learning (FL) enables collaborative model training without sharing raw data, which is attractive for network security analytics where traffic telemetry is sensitive. In this paper, we present a complete simulation study of Federated Averaging (FedAvg) for multi-class intrusion / traffic classification on the CIC–UNSW-NB15 augmented dataset. A multilayer perceptron (MLP) is trained across simulated clients that hold disjoint training shards; the server aggregates client weights by sample-count-weighted averaging. We evaluate a grid of 64 experimental configurations spanning client counts, global rounds, local epochs, and label-poisoning settings. Results show that longer training improves accuracy in clean settings, while targeted label poisoning is most damaging when a single client controls training (accuracy dropping to roughly 0.85–0.87). With multiple clients, FedAvg dilutes one poisoned client's influence and accuracy typically remains above 0.90. Confusion-matrix analysis further shows that targeted poisoning toward the majority class inflates false positives for that class and reduces recall for several minority attack types.

***Keywords*—** Federated learning, FedAvg, intrusion detection, data poisoning, CIC–UNSW-NB15, neural networks, confusion matrix

I. INTRODUCTION

Modern intrusion detection systems rely on large volumes of flow-level telemetry. Organizations are often unwilling to centralize raw traffic features due to privacy and compliance constraints. *Federated learning* addresses this tension by keeping data local while sharing only model updates.

This work studies a practical FL simulation for tabular network traffic classification on CIC–UNSW-NB15 (76 features, 10 classes), using a shared MLP and FedAvg aggregation, under a label-poisoning threat model on one client.

Contributions: (1) an end-to-end FedAvg pipeline; (2) a 64-run experimental sweep; (3) confusion-matrix / TP–FP–FN–TN analysis of poisoning effects; (4) public interactive dashboards for transparent inspection.

II. RELATED WORK

McMahan et al. introduced FedAvg for communication-efficient distributed SGD with periodic averaging. Surveys highlight FL challenges including non-IID data and adversarial clients. FL has been explored for collaborative intrusion detection, while poisoning and backdoor attacks show that malicious updates can corrupt global models if not filtered. Our study focuses on label flipping on one client and measures whether vanilla FedAvg already provides dilution when honest clients participate.

III. METHODOLOGY

A. Dataset and Preprocessing

Feature matrix X (N×76) comes from Dataset.csv; labels $y \in \{0,\ldots,9\}$ from Label.csv. Non-finite values are zeroed. A stratified 80/20 train/test split preserves class proportions. A StandardScaler is fit on training features only.

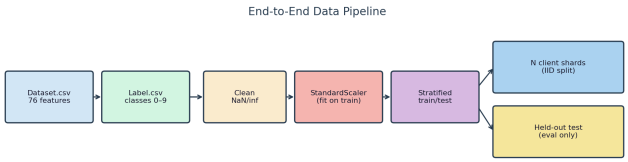


Fig. 1. End-to-end data pipeline from raw CSVs to client shards and a held-out test set.

B. Client Simulation

Training indices are shuffled with a fixed seed and split into N approximately equal shards (IID-like). Each shard never observes another client's raw rows. The test set is used only for global evaluation.

C. Model Architecture

All clients share an identical MLP: 76 → 128 (ReLU) → 128 (ReLU) → 10 (logits). Training uses Adam (lr = 10^{-3}) and cross-entropy loss.

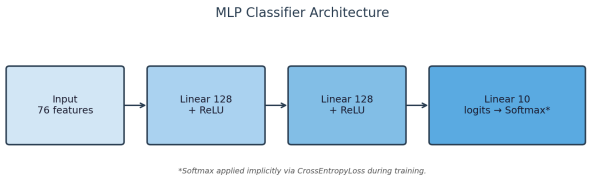


Fig. 2. Shared MLP classifier used by every client.

D. Federated Averaging

In each global round, the server broadcasts θ, each client trains locally for E epochs, then the server aggregates

$$\theta \leftarrow \Sigma_k w_k \theta_k, \quad w_k = n_k / \Sigma_j n_j$$

where n_k is the local sample count. The updated global model is evaluated on the held-out test set.

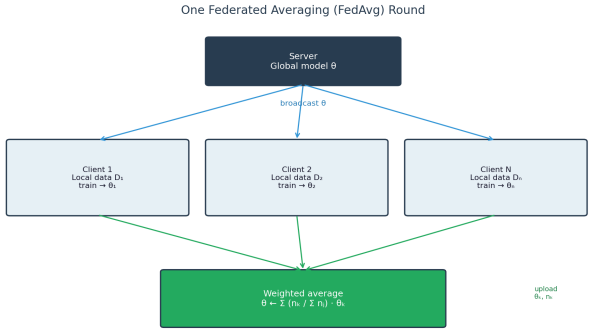


Fig. 3. One FedAvg round: broadcast, local training, and weighted aggregation.

E. Label Poisoning Attack

A fraction f of labels on client 0 are corrupted before training. Modes: *target* (flip to class 0), *random_wrong*, and *cycle*. Test labels remain clean.

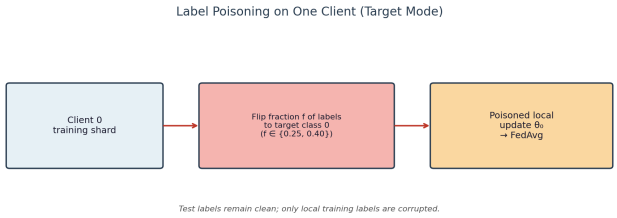


Fig. 4. Targeted label poisoning applied only to client 0 training labels.

IV. EXPERIMENTAL SETUP

We vary clients N ∈ {1,2,3,5}, rounds, local epochs E ∈ {1,2}, poison fraction f ∈ {0, 0.25, 0.40}, and poison mode. Metrics: accuracy, precision, recall, F1, plus confusion matrices for selected runs. Interactive results: https://fl-research.vercel.app.

V. RESULTS

A. Overall Sweep Findings

Across 64 runs: (1) longer training improves clean accuracy; (2) single-client 40% target poisoning is most damaging (≈0.85–0.87 accuracy); (3) with N ≥ 2, FedAvg dilutes one poisoned client and accuracy often remains ≥ 0.91.

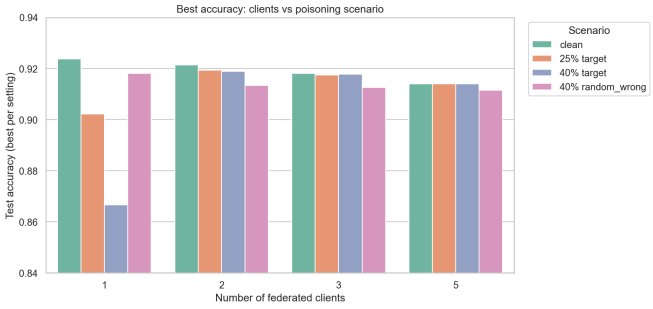


Fig. 5. Best accuracy by client count and poisoning scenario.

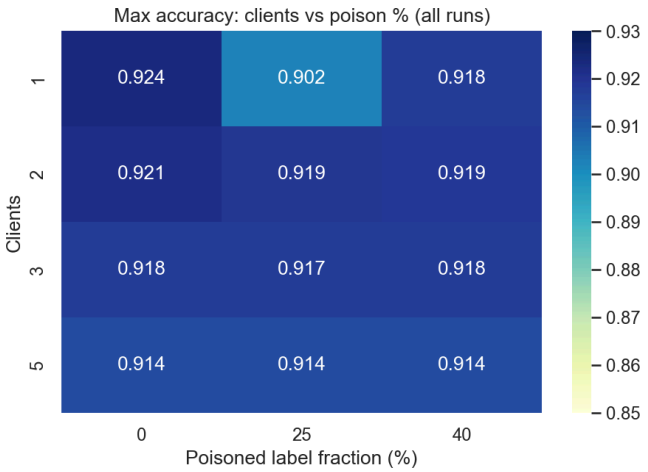


Fig. 6. Max accuracy heatmap: clients vs poison percentage.

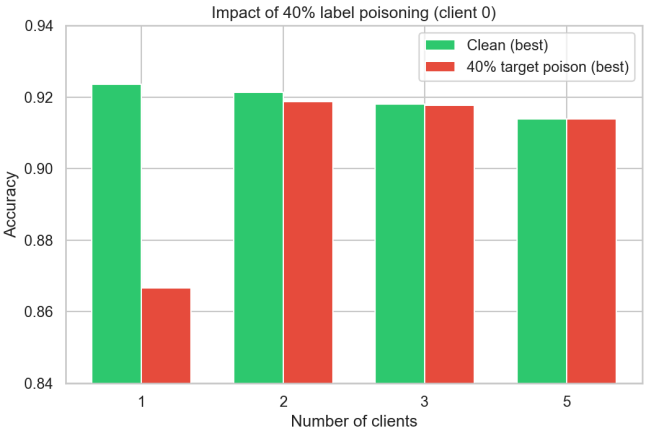


Fig. 7. Clean versus 40% target poisoning (best accuracy per client count).

TABLE I. Best test accuracy by client count and poison setting

Clients	Poison	Mode	Acc.	F1
1	0%	—	0.9237	0.9212
1	25%	target	0.9022	0.8881
1	40%	target	0.8666	0.8329
1	40%	random	0.9180	0.9127
2	0%	—	0.9214	0.9177
2	40%	target	0.9188	0.9132
3	0%	—	0.9181	0.9152
3	40%	target	0.9178	0.9135
5	0%	—	0.9139	0.9119
5	40%	target	0.9139	0.9117

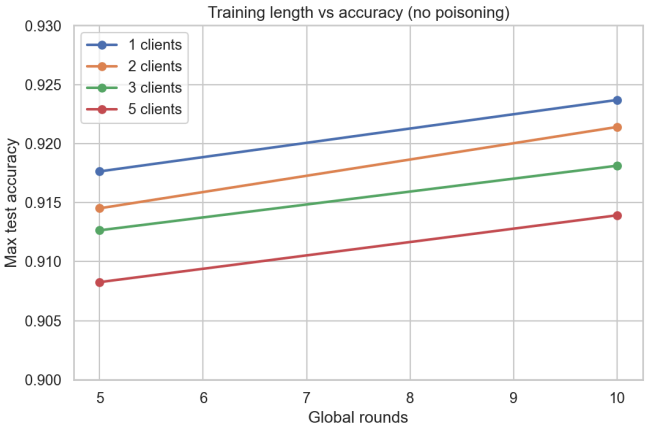


Fig. 8. Effect of global rounds on clean-run accuracy.

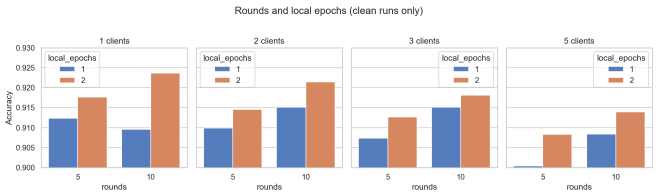


Fig. 9. Local-epoch effects for clean federated runs.

B. Confusion-Matrix Analysis

Class 0 dominates the test distribution. Clean multi-client runs achieve high class-0 recall (≈0.98) but miss rare classes. Many errors flow toward class 5. Under single-client 40% target poisoning, the model over-predicts class 0: large off-diagonal mass appears in column 0 for true classes 4–7, raising class-0 false positives and lowering minority-class recall.

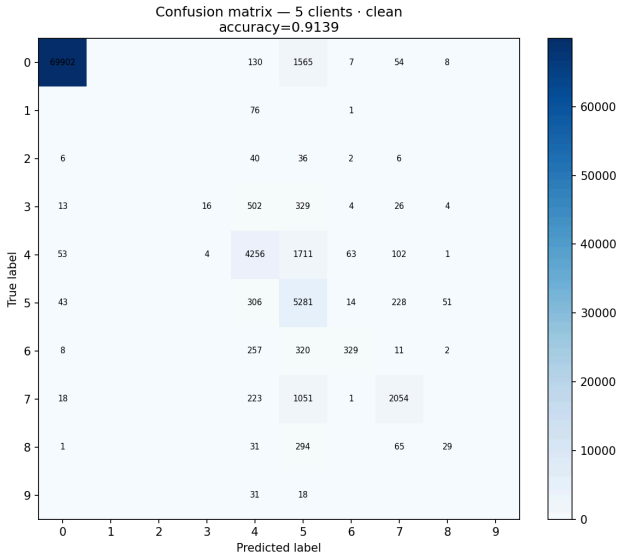


Fig. 10. Confusion matrix: 5 clients, clean.

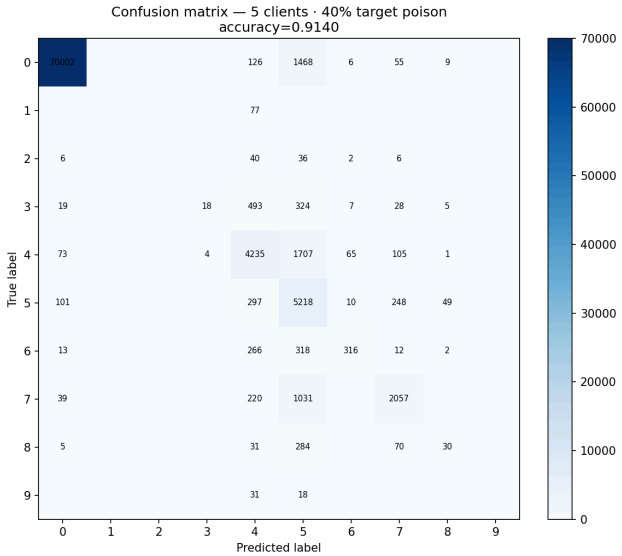


Fig. 11. Confusion matrix: 5 clients, 40% target poison.

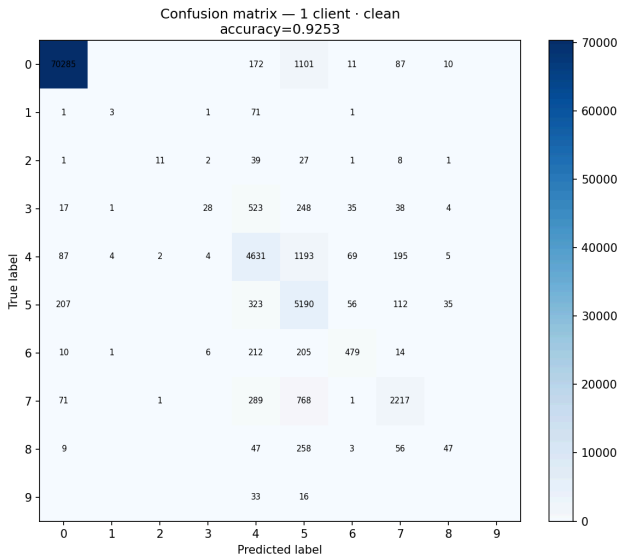


Fig. 12. Confusion matrix: 1 client, clean.

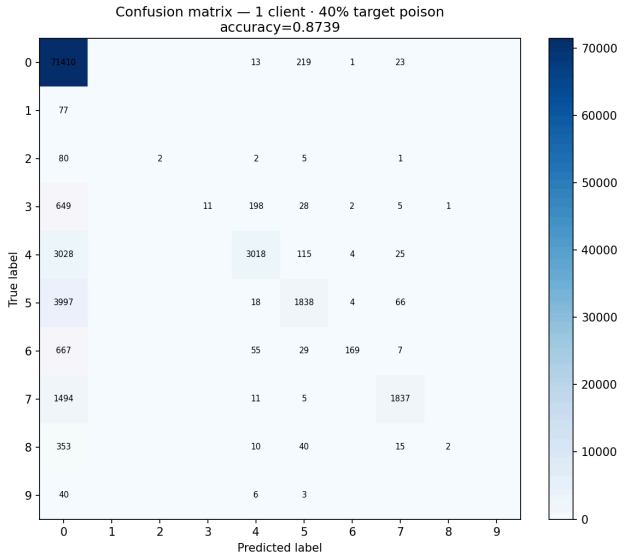


Fig. 13. Confusion matrix: 1 client, 40% target poison.

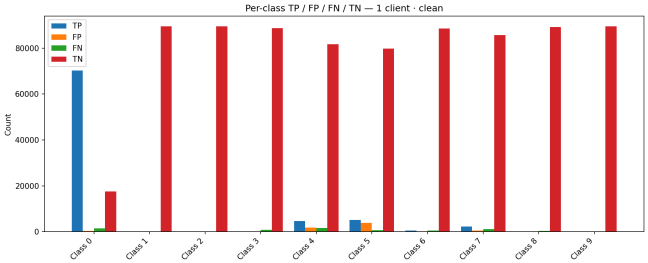


Fig. 14. Per-class TP/FP/FN/TN: 1 client, clean.



Fig. 15. Per-class TP/FP/FN/TN: 1 client, 40% target poison.

VI. DISCUSSION

Why multi-client FL resists this attack. Equal-sized shards imply w_k=1/N. Honest clients pull the average toward a better optimum.

Why single-client FL fails. Poisoned labels fully determine the decision boundary and encourage predicting the majority class when uncertain.

Limitations. IID shards, pooled scaler fitting, and a single MLP architecture. Non-IID partitions, robust aggregators, and differential privacy are future work.

VII. CONCLUSION

We presented an IEEE-style experimental study of FedAvg for CIC–UNSW-NB15 under clean and poisoned conditions. Sufficient training helps clean accuracy; targeted poisoning hurts most with one client; vanilla FedAvg dilutes one poisoned client when multiple clients participate; and confusion analysis is essential under class imbalance.

ACKNOWLEDGMENT

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